

NENA

Bidirectional Sign-to-Speech Mobile Platform

MVP Business Requirements Document & Project Management Plan

Based on BABOK® v3 (CBAP) · PMBOK® 7th Edition

Document Version: 1.0

Status: Approved for Development

Date: June 2026

Classification: Confidential — Internal Use Only

Target Release: MVP Q4 2026

East Africa Assistive Technology Initiative

Empowering the Deaf & Hard-of-Hearing Community

Document Control

Field	Detail
Document Title	NENA MVP Business Requirements & Project Management Plan
Version	1.0
Prepared By	NENA Project Team
Review Authority	Project Sponsor / Steering Committee
Frameworks Used	BABOK® v3 (CBAP), PMBOK® 7th Ed., Agile/Scrum
Methodology	Hybrid (Predictive + Agile)

Revision History

Version	Date	Author	Change Summary	Status
0.1	May 2026	Business Analyst	Initial draft from concept note	Draft
1.0	June 2026	BA Lead	Full MVP scope, risks, WBS, backlog added	Approved

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1. Executive Summary

NENA is an AI-powered bidirectional mobile communication platform designed to eliminate the communication barriers experienced by Deaf and hard-of-hearing individuals across East Africa. The platform enables **real-time, two-way translation** between Tanzanian/Kenyan Sign Language and spoken Swahili — both in face-to-face settings and during live VoIP telephone calls — without requiring a human interpreter.

Strategic Value Statement

NENA converts any smartphone into a pocket interpreter. It removes structural exclusion from healthcare, banking, and civic services for millions of Deaf citizens in East Africa by combining Few-Shot CLIP gesture recognition, OpenAI Whisper ASR, Neural TTS, and a WebRTC/SIP live calling engine inside a single Flutter mobile application.

Key MVP Outcomes

- A functional Flutter mobile app translating sign language to spoken Swahili audio and vice versa.
- An in-app VoIP calling system (Nena Call) enabling real-time relay between a Deaf and a hearing caller.
- A user voice-profile setting (Male/Female neural TTS) that respects the user's gender identity.
- Operations over constrained 3G networks via edge processing and lightweight token streaming.

2. Problem Statement & Business Need

2.1 Problem Analysis (BABOK — Business Need Elicitation)

The following root causes have been identified through stakeholder analysis and secondary research:

Root Cause	Business Impact
No affordable inter-preter access	Deaf individuals cannot independently access hospitals, banks, or government offices.
Complete telephonic exclusion	Standard GSM networks carry only audio; Deaf users cannot make or receive calls without a hearing proxy, destroying privacy.
Low-resource language data scarcity	TSL and KSL have minimal annotated corpora, making standard deep-learning impossible without a few-shot approach.
Swahili gender-neutral morphology	Pronouns like “Yeye” map to both He/She, breaking downstream TTS voice assignment without an explicit user profile override.

2.2 Opportunity Statement

The smartphone penetration rate in East Africa is growing rapidly, making a mobile-first solution the highest-impact delivery channel. By combining publicly available AI models (CLIP, Whisper) with edge processing, NENA can deliver professional-grade interpretation at near-zero marginal cost per user.

3. Project Scope

3.1 In-Scope for MVP

- **Sign-to-Speech (Outbound):** Real-time capture of sign gestures via front camera → Few-Shot CLIP classification → Swahili text → Neural TTS audio output (male/female profile).
- **Speech-to-Sign (Inbound):** Audio capture → Whisper ASR → Swahili text → 3D avatar sign animation on screen.
- **Nena Call (VoIP Relay):** WebRTC + SIP gateway enabling a live call between a Deaf user and a hearing caller with bidirectional translation in real time.
- **User Profile Settings:** Male/Female TTS voice profile, configurable at onboarding and in settings.
- **Core App Shell:** Flutter application (Android-first), onboarding screens, permissions handling, and offline-capable few-shot gesture dictionary.
- **Swahili Sign Dictionary — MVP Vocabulary:** A curated set of at minimum 100 high-frequency signs covering health, finance, and daily life.

3.2 Out of Scope for MVP

- × iOS version (Phase 2 release).
- × Full KSL/TSL corpus expansion beyond MVP vocabulary set.
- × Group/multi-party calling.
- × Integration with government or bank APIs.
- × Offline Whisper ASR (cloud API used for MVP).

4. Stakeholder Register (BABOK — Stakeholder Analysis)

Stakeholder		Role/Category	Interest / Influence	Engagement Strategy
Deaf End Users		Primary Beneficiary	High interest; limited technical influence. Core UX validators.	Field testing; focus groups; user story validation.
Hearing Callers		Secondary User	Will use the VoIP call relay; need zero learning curve.	Usability tests on hearing side of call.
Sign Language Interpreters		SME / Advisor	Authority on TSL/KSL accuracy; potential threat if not involved.	Co-design gesture dictionary; community advisory board.
Project Sponsor	Sponsor	Funder / Governance	High influence; approves budget and scope.	Weekly status reports; milestone demos.
Flutter Team	Dev	Delivery	Builds mobile app; needs clear API contracts.	Sprint planning; daily standups; detailed API specs.
AI/ML neer	Engi-	Delivery	Owns CLIP & Whisper integration; needs data and compute.	Bi-weekly technical syncs; shared model registry.
VoIP / Engineer	Infra	Delivery	Owns WebRTC, SIP, and Twilio integration.	Architecture review; API contract sign-off.
Deaf community (NGOs)	Com- Orgs	Community Partner	High influence on trust and adoption; key data-sharing channel.	MOU for co-design; data consent agreements.

Stakeholder	Role/Category	Interest / Influence	Engagement Strategy
Regulatory/Privacy Body	Compliance	Biometric / video data handling oversight.	Privacy impact assessment; GDPR/local law review.

5. Business Requirements (BABOK — Requirements Classification)

Requirements are classified using the BABOK Knowledge Area taxonomy.

5.1 Business Requirements (WHY)

BR — Business Requirements

- BR-1.** NENA shall eliminate the need for third-party human interpreters for Deaf users conducting common daily transactions (medical, financial, civic).
- BR-1.** NENA shall enable Deaf individuals to independently initiate and receive telephone calls.
- BR-1.** NENA shall operate on devices with 3G connectivity and mid-range Android hardware (2019 or newer).
- BR-1.** NENA shall preserve the gender identity and vocal dignity of Deaf users through a configurable TTS voice profile.

5.2 Stakeholder Requirements (WHO)

SR — Stakeholder Requirements

- SR-2.** Deaf users shall be able to perform sign language in front of the phone camera and hear their words spoken aloud within 2 seconds.
- SR-2.** Deaf users shall be able to read incoming speech as sign language animations on screen in near real-time.
- SR-2.** Deaf users shall be able to select Male or Female TTS voice during onboarding and change it at any time in Settings.
- SR-2.** Hearing callers shall experience the call as a standard voice call with no special app required on their end.

SR-2. The sign language dictionary shall include vocabulary validated by the regional Deaf community.

5.3 Solution Requirements — Functional (WHAT)

5.3.1 Sign-to-Speech Module

ID	Requirement	Priority	Status	Acceptance Criterion
FR-01	App shall access the front-facing camera at minimum 24 fps for gesture capture.	Must Have	Open	Camera feed confirmed at ≥ 24 fps on target device.
FR-02	CLIP model shall classify a gesture from a 5-frame sequence within 1.5 seconds.	Must Have	Open	P90 latency ≤ 1.5 s on Snapdragon 665.
FR-03	Classified gesture shall map to a Swahili text token and display on screen.	Must Have	Open	Correct text token rendered for $\geq 90\%$ of calibrated signs.
FR-04	Swahili text shall be passed to TTS engine and spoken aloud through the device speaker/earpiece.	Must Have	Open	TTS audio produced in ≤ 500 ms from token.
FR-05	TTS shall use the user's selected voice profile (male/female).	Must Have	Open	Audio matches selected profile in blind test.
FR-06	Gesture dictionary shall cover ≥ 100 validated Swahili signs at MVP launch.	Must Have	Open	Dictionary file contains 100+ entries, community-reviewed.

5.3.2 Speech-to-Sign Module

ID	Requirement	Priority	Status	Acceptance Criterion
FR-07	App shall capture microphone audio and pass to Whisper ASR API.	Must Have	Open	ASR call initiated within 300 ms of voice end-detection.
FR-08	Whisper shall return Swahili text with $\geq 85\%$ word accuracy for clear speech.	Must Have	Open	WER $\leq 15\%$ in structured UAT with Swahili speakers.
FR-09	Text shall be semantically reduced to structural keywords before sign rendering.	Must Have	Open	Rendered animation is grammatically valid TSL/KSL sequence.
FR-10	3D avatar shall render the sign animation sequence within 1 second of text receipt.	Must Have	Open	Animation rendered ≤ 1 s on target device.

5.3.3 Nena Call — VoIP Relay Module

ID	Requirement	Priority	Status	Acceptance Criterion
FR-11	App shall initiate a VoIP call via WebRTC peer connection through the SIP/Twilio gateway.	Must Have	Open	Successful call setup ≤ 5 s on 3G.
FR-12	Incoming audio from hearing caller shall be routed to Whisper ASR and displayed as sign animation in real time.	Must Have	Open	End-to-end latency (voice $\rightarrow$ animation) ≤ 3 s.

ID	Requirement	Priority	Status	Acceptance Criterion
FR-13	Deaf user's front-camera gesture stream shall be processed on-device (no video upload).	Must Have	Open	No video bytes transmitted to cloud server during call.
FR-14	Recognised text tokens shall be transmitted to cloud and converted to TTS audio injected into the hearing caller's phone call.	Must Have	Open	Hearing caller receives audible Swahili speech.
FR-15	Call shall remain stable over a 3G connection (≥ 1 Mbps).	Must Have	Open	Zero dropped calls in 10-minute session over 3G sim.

5.3.4 User Profile & Settings Module

ID	Requirement	Priority	Status	Acceptance Criterion
FR-16	Onboarding flow shall include a mandatory voice profile selection screen (Male / Female).	Must Have	Open	100% of new users encounter profile screen.
FR-17	Voice profile selection shall persist in local device storage and survive app restart.	Must Have	Open	Profile unchanged after force-close and relaunch.
FR-18	User shall be able to change voice profile at any time from the Settings screen.	Must Have	Open	Profile change reflected in next TTS output.

5.4 Solution Requirements — Non-Functional (QUALITY)

ID	Category	Requirement	Measure
NFR-01	Performance	Sign classification latency ≤ 1.5 s P90.	Profiling on Snapdragon 665, 3GB RAM.
NFR-02	Performance	TTS audio output ≤ 500 ms from token.	Timestamp logging in test harness.
NFR-03	Availability	App uptime $\geq 99\%$ excluding third-party API downtime.	Crash reporting via Firebase Crashlytics.
NFR-04	Bandwidth	Call session data ≤ 200 kbps (tokens only, no video upload).	Network monitor during 10-min test call.
NFR-05	Privacy	No video frames stored or transmitted to server during a call.	Code audit + network packet inspection.
NFR-06	Accessibility	All UI controls navigable with a single hand.	Ergonomics review with 5 Deaf testers.
NFR-07	Compatibility	Android 9.0 (API 28) and above.	CI test matrix on Android 9, 11, 13, 14.
NFR-08	Security	User profile data encrypted at rest (AES-256).	Security audit before public release.
NFR-09	Usability	New Deaf user completes onboarding in ≤ 3 minutes.	Observed time-on-task in UAT session.

6. Solution Architecture Overview

6.1 System Pipeline Summary

Stream A — Outbound (Sign $\rightarrow$ Speech):

Camera (24+ fps) $\rightarrow$ Frame Buffer (5 frames) $\rightarrow$ CLIP Few-Shot Classifier (on-device, ONNX) $\rightarrow$ Swahili Text Token $\rightarrow$ Profile Router (Male/Female) $\rightarrow$ Neural TTS Engine $\rightarrow$ Speaker / VoIP Payload

Stream B — Inbound (Speech → Sign):

Microphone / VoIP Audio Packet → Whisper ASR (Cloud API) → Swahili Text → Semantic Keyword Reducer → Sign Dictionary Lookup → 3D Avatar Renderer (on-device)

6.2 Architecture Layer Responsibilities

Layer	Responsibility	Owner / Technology
Media Capture	Camera access, mic permissions, frame buffering	Flutter Camera Plugin / WebRTC MediaDevices
Outbound Vision Engine	Few-shot CLIP gesture classification, token output	AI/ML Team — ONNX Mobile / PyTorch Lite
Voice Routing	Profile state management (Male/Female), TTS call dispatch	Flutter Dev — BLoC / Dart State Provider
Inbound Audio Parser	Whisper ASR, keyword extraction	AI/ML Team — OpenAI Whisper API
Animation Renderer	Sign dictionary reference, 3D skeletal avatar display	Flutter Dev — Unity Embedded Widget / WebGL
Telephony Bridge	WebRTC session, SIP signalling, Twilio gateway, PCM routing	VoIP/Infra Team — Twilio Voice SDK, SIP
Backend Cloud	Text token relay, TTS audio synthesis, ASR proxy	Backend Team — Node.js / GCP Cloud Run

7. MVP Product Backlog (Agile — User Stories)

User stories follow the standard INVEST criteria. Priority uses MoSCoW notation.

7.1 Epic 1: Sign-to-Speech Translation

ID	User Story	Priority	Points	Sprint
US-01	As a <i>Deaf user</i> , I want to open the app and see my camera feed so that I can begin signing immediately.	Must Have	3	1
US-02	As a <i>Deaf user</i> , I want the app to recognise my sign and display the Swahili word on screen within 2 seconds so I can confirm the translation.	Must Have	8	2
US-03	As a <i>Deaf user</i> , I want the translated word to be spoken aloud so the person in front of me can hear it.	Must Have	5	2
US-04	As a <i>Deaf user</i> , I want to sign a full sentence and have it spoken as a complete phrase so my communication sounds natural.	Should Have	8	3
US-05	As an <i>admin/BA</i> , I want to add new sign entries to the dictionary so the system can be expanded without retraining the full model.	Must Have	5	2

7.2 Epic 2: Speech-to-Sign Translation

ID	User Story	Priority	Points	Sprint
US-06	As a <i>Deaf user</i> , I want to see an avatar sign the words a hearing person speaks so I can understand them without needing to lip-read.	Must Have	8	3
US-07	As a <i>Deaf user</i> , I want the avatar to respond in under 2 seconds of the speaker finishing their sentence.	Must Have	5	3
US-08	As a <i>hearing person</i> , I want to speak naturally in Swahili without pressing any button so the interaction feels seamless.	Should Have	3	3

7.3 Epic 3: Nena Call — VoIP Relay

ID	User Story	Priority	Points	Sprint
US-09	As a <i>Deaf user</i> , I want to tap a “Nena Call” button to call any phone number and have my signs automatically relayed as speech to the other person.	Must Have	13	4
US-10	As a <i>hearing caller</i> , I want to call a Deaf user’s Nena number from my regular phone and hear their response as natural Swahili.	Must Have	8	4
US-11	As a <i>Deaf user on a call</i> , I want to see the hearing caller’s speech as sign animations in real time so I can follow the conversation.	Must Have	13	5
US-12	As a <i>Deaf user</i> , I want the call to work over a basic 3G data connection so I can use it in rural or low-coverage areas.	Must Have	5	5

7.4 Epic 4: User Identity & Onboarding

ID	User Story	Priority	Points	Sprint
US-13	As a <i>new Deaf user</i> , I want to select my preferred voice (Male or Female) during onboarding so the app speaks with my voice identity.	Must Have	3	1
US-14	As a <i>Deaf user</i> , I want to change my voice profile in Settings at any time.	Must Have	2	1
US-15	As a <i>Deaf user</i> , I want the onboarding to show me how to sign 5 common words so I understand how the camera needs to see my hands.	Should Have	3	1

8. Work Breakdown Structure (PMBOK — WBS)

The WBS is structured to 3 levels across 4 project phases aligned with the concept note milestones.

WBS ₇	Phase / Deliverable	Work Package	Responsible Team
1.0 — Phase 1: AI & Data Foundation			
1.1	Sign Dictionary	Collect & validate 100+ TSL/KSL video samples	AI/ML + Community NGO
1.2	CLIP Model	Fine-tune few-shot CLIP on calibration keyframes; export ONNX	AI/ML Team
1.3	Whisper Integration	Configure Whisper API endpoint; build Swahili accuracy benchmarks	AI/ML Team
1.4	Semantic Reducer	Build keyword extraction layer (stopword removal, TSL grammar mapping)	AI/ML Team
1.5	Backend Service	Deploy cloud token relay API (Node.js / GCP)	Backend Team
2.0 — Phase 2: Mobile App — Core Features			
2.1	App Shell	Flutter project setup, routing, permissions, state management (BLoC)	Flutter Dev
2.2	Camera Module	Real-time camera feed, frame capture, ONNX model inference runner	Flutter Dev + AI/ML
2.3	TTS Module	Neural TTS integration, Male/Female profile wiring, audio output	Flutter Dev
2.4	Avatar Module	3D avatar renderer integration, sign animation playback	Flutter Dev
2.5	Onboarding & Settings	Voice profile selection, onboarding tutorial screens, settings page	Flutter Dev
3.0 — Phase 3: Nena Call — VoIP Engine			
3.1	WebRTC Client	Peer connection setup, ICE/STUN/TURN configuration	VoIP/Infra Team

WBS ₇	Phase / Deliverable	Work Package	Responsible Team
3.2	SIP Gateway	Twilio Programmable Voice SIP bridge; phone network inbound/outbound	VoIP/Infra Team
3.3	Stream Split Logic	Concurrent Stream A (gestures) and Stream B (audio) during live call	VoIP + AI/ML
3.4	Edge Optimisation	Token-only cloud transmission; bandwidth profiling; 3G test	All Teams
4.0 — Phase 4: Testing, Validation & Launch			
4.1	Unit & Integration Testing	Automated test suite for all modules; CI/CD pipeline (GitHub Actions)	All Developers
4.2	Community UAT	Field testing with Deaf community NGO; focus groups; sign accuracy review	BA + Community
4.3	Performance Profiling	Latency benchmarks on target devices; 3G network simulation	QA Team
4.4	Privacy Audit	Confirm no video frames transmitted; data at rest encryption check	Security Lead
4.5	MVP Release	Android APK to closed beta; Play Store submission	Flutter Dev + PM

9. Project Schedule — MVP Roadmap

Target: **MVP Release in Q4 2026** (6 sprint cycles × 2 weeks).

Sprint	Dates (est.)	Goals / Stories	MVP Gate
S1	Wk 1–2	App shell; onboarding; voice profile settings (US-01, US-13, US-14, US-15)	App launches & profile saved
S2	Wk 3–4	CLIP model on-device; sign-to-text; TTS output (US-02, US-03, US-05)	Single sign spoken aloud
S3	Wk 5–6	Whisper ASR; keyword reducer; avatar renderer; phrase signing (US-04, US-06, US-07, US-08)	Full bidirectional face-to-face mode
S4	Wk 7–8	WebRTC client; SIP/Twilio gateway; basic VoIP call (US-09, US-10)	First live relay call placed
S5	Wk 9–10	Bidirectional call (gestures + incoming speech in call); 3G optimisation (US-11, US-12)	Full Nena Call working end-to-end
S6	Wk 11–12	UAT; performance profiling; bug fixes; privacy audit; beta release	MVP APK shipped to beta testers

10. Risk Register (PMBOK — Risk Management)

ID	Risk	Likelihood	Impact	Rating	Mitigation Strategy
R-01	CLIP accuracy insufficient for real-world TSL gesture variation	High	High	Critical	Increase calibration samples per sign; allow user re-calibration; set 85% accuracy threshold as Sprint 2 gate.

ID	Risk	Likelihood	Impact	Rating	Mitigation Strategy
R-02	Whisper ASR accuracy degrades on accented Swahili or noisy environments	Medium	High	High	Fine-tune Whisper on Tanzanian Swahili corpus; add noise-cancellation pre-processing layer.
R-03	VoIP call latency >3 s on 3G makes call unusable	Medium	High	High	Edge-only video processing (no upload); token-only cloud path; load test on 3G sim before S5 gate.
R-04	Deaf community does not trust or adopt the app	Medium	High	High	Co-design dictionary with community NGO from Sprint 1; involve interpreters as advisors; run trust workshops.
R-05	Twilio / SIP integration more complex than estimated	Medium	Medium	Medium	Spike research in Sprint 3 before full S4 build; hold 3 days buffer per sprint.
R-06	Privacy concerns about front-camera usage	Low	High	High	Publish privacy notice; code audit confirms zero video upload; pursue GDPR-aligned data consent flow.
R-07	Key AI/ML developer unavailable mid-project	Low	High	Medium	Document model architecture fully; cross-train one Flutter dev on ONNX inference.

ID	Risk	Likelihood	Impact	Rating	Mitigation Strategy
R-08	Swahili sign dictionary too small for real-world utility	Medium	Medium	Medium	Fast-path 100 highest-frequency signs for MVP; design extensible dictionary schema from Sprint 1.

11. Team Roles & RACI Matrix (PMBOK — Resource Management)

Work Package	Stakeholder					
	PM	BA	AI/ML	Flutter	VoIP	QA
Sign Dictionary Collection	A	R	C	I	I	I
CLIP Model Development	I	C	R	C	I	A
Whisper ASR Integration	I	C	R	C	I	A
Flutter App Shell	A	C	I	R	I	I
Camera / Inference Module	A	I	C	R	I	C
TTS & Voice Profile	A	C	C	R	I	I
Avatar Renderer	A	I	C	R	I	I
WebRTC Client	A	I	I	C	R	C
SIP / Twilio Bridge	A	I	I	C	R	C
Stream Split (VoIP + AI)	A	I	C	C	R	C
UAT & Community Testing	I	R	C	C	C	A
Privacy & Security Audit	A	C	I	I	I	R
MVP Release	R	C	I	C	I	A

R = Responsible **A** = Accountable **C** = Consulted **I** = Informed

12. Definition of Done & MVP Acceptance Criteria

The MVP is considered **DONE** and ready for community beta release when ALL of the following criteria are verified:

#	Criterion	Verification Method	Met?
1	Sign-to-Speech works	≥90% of 100 dictionary signs correctly translated & spoken in UAT.	
2	Speech-to-Sign works	Avatar renders correct sign for ≥85% of Swahili sentences in UAT.	
3	Nena Call end-to-end	Live call between Deaf user and hearing caller completed successfully with bidirectional relay.	
4	Voice profile works	Male and Female TTS profiles produce correct gendered audio; persists across app restarts.	
5	Performance targets met	P90 sign latency ≤ 1.5 s; call latency ≤ 3 s; stable on 3G.	
6	Privacy audit passed	Zero video frames transmitted to server confirmed by packet inspection.	
7	Community validation	At least 10 Deaf users complete 15-min usability session with ≥80% task success rate.	
8	No critical/high bugs	Zero P1 bugs open; all P2 bugs have accepted workarounds.	

13. Team Onboarding Guide — How to Use This Document

Each sub-team should use this document as their primary reference. The table below maps each team to the sections most relevant to their work.

Team	Primary Sections to Read & Act On
All Teams	Section 1 (Vision), Section 2 (Problem), Section 3 (Scope), Section 10 (Definition of Done).
Flutter Developers	Section 5 (FR-01 to FR-18, NFR-01 to NFR-09), Section 6 (Architecture), Section 7 (Epics 1–4), Section 8 (WBS 2.0), Section 9 (Sprint Plan).
AI / ML Engineers	Section 5 (FR-01 to FR-10), Section 6 (Outbound & Inbound pipelines), Section 7 (Epics 1–2), Section 8 (WBS 1.0).
VoIP / Infra Engineers	Section 5 (FR-11 to FR-15, NFR-04 to NFR-05), Section 6 (Telephony Bridge layer), Section 7 (Epic 3), Section 8 (WBS 3.0).
QA Engineers	Section 5 (all FRs & NFRs, Acceptance Criteria columns), Section 10 (Definition of Done), Section 8 (WBS 4.0).
Business Analyst	All sections. Owns requirements baseline, user story grooming, stakeholder communication, and UAT coordination.
Project Manager	Section 4 (Stakeholders), Section 8 (WBS), Section 9 (Schedule), Section 10 (DoD), Section 11 (RACI), Appendix (Risks).

Daily Development Principle

At the start of every sprint, any developer picking up a user story should: (1) read the linked functional requirement; (2) confirm the acceptance criterion is understood and testable; (3) raise any ambiguity with the BA before writing a single line of code. Stories are not “done” until the acceptance criterion passes in a demo.

14. Communication Plan (PMBOK — Communications Management)

Meeting / Artefact	Purpose	Audience	Frequency	Owner
Daily Standup	Blockers, progress, plan for today	Dev Teams	Daily (15 min)	Scrum Master
Sprint Planning	Story commitment; acceptance criterion review	All Teams	Bi-weekly	PM + BA
Sprint Review / Demo	Show working software to stakeholders	All + Sponsor	Bi-weekly	PM
Sprint Retrospective	Process improvement	Dev Teams	Bi-weekly	Scrum Master
Technical Architecture Sync	API contracts; model interface alignment	All Dev Teams	Weekly	Tech Lead
Community UAT Session	Field testing with Deaf users	BA + NGO Partner	Sprint 6	BA
Sponsor Status Report	RAG status; risks; milestone gate	Project Sponsor	Weekly	PM

15. Assumptions & Constraints

Assumptions

- The community NGO partner will facilitate access to at least 20 Deaf participants for dictionary video collection within Sprint 1.
- OpenAI Whisper API has acceptable uptime and latency for Swahili during the project period.
- Twilio Programmable Voice supports SIP-based mobile inbound calls in Tanzania/Kenya at acceptable cost.
- The Flutter team has prior Dart/Flutter experience; no framework ramp-up time is budgeted.

- A device running Android 9.0+ with a functional front camera is available to the Deaf user.

Constraints

- MVP must operate on 3G connectivity (minimum ~1 Mbps); no assumption of 4G/LTE.
- No video data may be stored or uploaded to the server (privacy constraint, non-negotiable).
- MVP is Android-only; iOS is explicitly deferred post-MVP.
- MVP sign dictionary capped at 100 signs for time and data reasons; expansion is a Phase 2 item.
- All AI model inference for the outbound pipeline must run on-device to preserve call stability and user privacy.

16. Glossary

Term	Definition
ASR	Automatic Speech Recognition — software that converts spoken audio to text.
BABOK®	Business Analysis Body of Knowledge — IIBA framework for business analysis practice.
BLoC	Business Logic Component — a Flutter state management pattern separating UI from business logic.
CLIP	Contrastive Language–Image Pretraining — OpenAI model aligning image and text embeddings.
Edge Processing	Running AI model inference on the local device rather than a remote server.
Few-Shot Learning	A machine learning paradigm enabling a model to generalise from very few training examples.
KSL	Kenyan Sign Language.
ONNX	Open Neural Network Exchange — a portable format for deploying ML models on mobile devices.
PMBOK®	Project Management Body of Knowledge — PMI framework for project management.
SIP	Session Initiation Protocol — a signalling standard for VoIP communications.

Term	Definition
TTS	Text-to-Speech — software that synthesises spoken audio from text input.
TSL	Tanzanian Sign Language.
UAT	User Acceptance Testing — validation by end users against acceptance criteria.
VoIP	Voice over Internet Protocol — phone calls carried over an internet data connection.
WebRTC	Web Real-Time Communication — open standard for browser/app-based real-time audio/video.
WER	Word Error Rate — the standard metric for ASR accuracy (lower is better).

— END OF DOCUMENT —

This document is the primary reference for all NENA MVP team members.

Any changes to requirements, scope, or architecture must go through the BA and PM for impact assessment and version control before implementation.

NENA Project · East Africa Assistive Technology Initiative · 2026